

UNDERSTANDING THE BANK LOAN DATASET

A Technical Data Dictionary & Domain Glossary for Credit Portfolio Analytics

Analytical Context & Business Case: In retail and corporate banking, managing loan portfolio performance is crucial for maintaining credit quality, assessing liquidity, and mitigating loan default risk. This data dictionary serves as a technical blueprint for the **Bank Loan Analytics Dashboard**. It documents all **23 data columns** comprising the **38,576 borrower records** in the credit portfolio. These columns enable the calculation of vital banking KPIs and support conditional credit classification in Power Query.

1. Core Portfolio Dataset Dictionary

Column Name	Data Type	Business Definition & Analytic Value
id	Whole Number	A unique, sequential identification number assigned to each borrower's loan application. Used to establish total application counts.
address_state	Text	The US state of residence provided by the borrower. Enables geospatial slicing to isolate regional economic concentrations.
annual_inc	Decimal	The self-reported annual income provided by the borrower during registration. Crucial for assessing loan repayment capacity.
dti	Percentage	Debt-to-Income (DTI) ratio, calculated as monthly debt payments (excluding mortgage and this loan) divided by borrower's monthly income.
earliest_cr_line	Date	The month and year the borrower's earliest reported credit line was opened. Serves as a measure of credit history length.
emp_length	Text	Employment length of the borrower in years (e.g., '< 1 year', '1 year', ..., '10+ years'). Acts as a proxy for job stability.
emp_title	Text	The job title or profession supplied by the borrower when applying for the credit line.
grade	Text	Overall risk grade assigned by the bank (A = Lowest risk, G = Highest risk). Higher grades correspond to higher creditworthiness.
sub_grade	Text	A granular risk sub-grade assigned to the loan (e.g., A1, A2, B1...), allowing banks to refine pricing and interest structures.
home_ownership	Text	The home ownership status reported by the borrower (e.g., RENT, MORTGAGE, OWN, OTHER). Helps assess collateral and stability.
installment	Decimal	The monthly payment amount owed by the borrower if the loan is approved and originates.
int_rate	Percentage	The annual interest rate charged on the issued loan. This represents the bank's primary revenue yield on funded capital.
issue_date	Date	The official date the loan was approved and capital disbursed to the borrower. Crucial for timeline-based trend analysis.

Column Name	Data Type	Business Definition & Analytic Value
loan_status	Text	The current operational status of the loan (Fully Paid, Current, Charged Off). Indicates whether the borrower is active or in default.
purpose	Text	The borrower's self-selected category explaining the reason for the loan request (e.g., Debt Consolidation, Home Improvement).
term	Text	The repayment duration of the loan, structured as either 36 months (3-year) or 60 months (5-year) maturities.
verification_status	Text	Indicates whether the borrower's self-reported income source was verified by the bank (Verified, Source Verified, Not Verified).
total_payment	Decimal	The total cumulative payment amount (principal + interest) repaid by the borrower back to the bank to date.
loan_amount	Decimal	The total principal loan amount approved and funded to the borrower at origination.
last_payment_date	Date	The date on which the bank received the most recent payment from the borrower.
last_credit_pull_d	Date	The most recent month the bank pulled the borrower's credit report to monitor ongoing risk.
total_rec_prncp	Decimal	The portion of the cumulative repayment amount that went toward reducing the loan's outstanding principal balance.
total_rec_int	Decimal	The portion of the cumulative repayment amount representing interest paid, reflecting direct interest revenue to the bank.

2. Power BI Transformation & KPIs

The columns described above do not merely reside in storage; they are actively transformed in **Power Query Editor** and modeled with **DAX** to feed the dashboard's analytics:

- **Good vs. Bad Loans:** Derived via a Conditional Column in Power Query based on the loan_status field. If loan_status is 'Fully Paid' or 'Current', the loan is classified as **Good**. If 'Charged Off', it is classified as **Bad** (defaulted). This drives the dashboard's core segmentation.
- **Total Loan Applications:** Calculated as COUNT(financial_loan[id]). ID is utilized because it represents a guaranteed unique, non-null applicant identifier.
- **Total Funded Amount:** Derived by summing loan_amount, which measures the aggregate capital disbursed.
- **Total Amount Received:** Sum of total_payment, measuring cash inflow to track institutional cash flow.
- **Geographical Standardization:** The address_state text column was specifically categorized as a **State** within Power BI to ensure that the bubble maps render state boundaries inside the United States accurately.